

CAREER OBJECTIVE

- Detail-oriented Computer Science undergraduate with strong skills in **software development, AI/ML, and full-stack engineering**. Passionate about solving real-world problems, building scalable applications, and exploring innovative technologies to deliver impactful solutions.

EDUCATION

Degree	Specialization	Institute	Year	CPI/%
B.Tech	Computer Science and Engineering (AI-ML)	Keshav Memorial Institute of Technology	2022-Present	8.54
HSC	Physics, Chemistry, Mathematics	Narayana Junior College	2020-2022	92.3%
SSC	-	Kendriya Vidyalaya Sangathan (CBSE)	2020	88.4%

TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript, C++, SQL
- Frameworks & Libraries:** TensorFlow, PyTorch, LangChain, HuggingFace Transformers
- Web Development:** Node.js, Express.js, Flask, Streamlit, React.js
- Databases:** MongoDB, MySQL
- Tools:** Docker, Git, Postman, Google Colab, Kaggle

PROJECTS

CheXNet with Attention-based Medical Report Generation TensorFlow | Bahdanau Attention | NLP
<https://github.com/Karthik3116/Medical-Report-Generator>

- Developed an AI system that converts chest X-ray images into descriptive reports using a CNN encoder and GRU-based RNN decoder.
- Trained on the NLMCXR dataset, achieving strong BLEU score performance.
- Built a web interface for X-ray upload and real-time report generation.

Smart PDF Chat LangChain | FAISS | Python | MongoDB | Streamlit
https://github.com/Karthik3116/RAG_BOT | <https://grobobot.streamlit.app/>

- Built a web app to upload PDFs and query them in natural language with **reference sources (page numbers, chunks, and links)** for each response.
- Implemented PDF chunking, embedding, and semantic search using FAISS.
- Integrated LangChain for retrieval + generation pipeline with session memory.
- Enhanced credibility by displaying **inline PDF references** alongside answers.
- Designed a Streamlit UI with chat interface and downloadable conversation history including references.

aiXinterview – Voice AI Interview Simulator MERN | Flask | Vapi
<https://github.com/Karthik3116/aiXinterview> | <https://interview.karthik.top/>

- Developed a voice-based AI platform for mock interviews with real-time feedback.
- Flask microservices handled question generation, scoring, and analytics.
- Implemented interactive UI for candidates to simulate real-world interviews.

EXTRA-CURRICULAR ACHIEVEMENTS/ACTIVITIES

- Developed a Video Summarizer Web App at CBIT Hacktoberfest Hackathon.
- Achieved Top 26% rank on LeetCode by solving algorithmic problems consistently.
- Finalist at HASCKAVVY Hackathon, built and pitched a brain tumor classification model for MRI scans.